

Response to Reviewers

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NISQ-Aware Quantum Kernel SVM for Network Intrusion Detection: A Regime-Specific Benchmark on NSL-KDD

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Dear Editor-in-Chief, Associate Editor, and Reviewers,

We thank the reviewers for reviews that were unusually specific and, in several places, correct about errors we had not found ourselves. Reviewer 4 identified a false theorem and a selection step that filters nothing; Reviewer 2 identified two citation defects, one of which was a reference we could not verify exists; Reviewer 1 identified an asymmetric tuning protocol that favoured our own method. Each of these was checked and each was confirmed.

Acting on them changed the paper’s conclusion. We ask the editors to read the following before the point-by-point responses, because it is the most important thing we have to report.

Five claims from the submitted version are withdrawn

Re-running every experiment under a corrected protocol — ten runs instead of five, symmetric hyper-parameter tuning, a random forest and XGBoost added as baselines, and results reported on the full KDDTest⁺ partition rather than a 300-sample subset — does not support five claims made in the submitted version. We withdraw them rather than qualify them.

Submitted version	Measured under the revised protocol
Advantage in the low-data regime	<i>Reversed.</i> Classical baselines win at small N ; the ordering flips between $N = 2000$ and $N = 5000$
“+6.7 points on the rare-attack subset, Cohen’s $d = +0.68$ ”	<i>Not reproducible.</i> No rare-class metric was computed by the code that produced it
Theorem 1: $n^* = 4$ maximises J	<i>False</i> , as Reviewer 4 states. J is maximised at $n = 2$ (0.551)
QSVM-ZZ 0.854 > SVM-RBF 0.838	With a tree ensemble in the comparison, XGBoost 0.8503 > QSVM-ZZ 0.8469 > random forest 0.8446, intervals overlapping
“Quantum advantage is real and measurable”	Of 110 controlled comparisons: 21 favour the quantum kernel, 21 favour a classical baseline, 68 are inconclusive

We also report three defects that no reviewer raised and that we found while preparing this revision. They are listed in Sec. 6 of this letter and stated in the manuscript itself, in the erratum subsection of Sec. III. One of them is an error in the single theoretical result the revision retains.

What replaces the withdrawn claims is not a weaker version of the same story. It is a different result: an ordering that *reverses* with training-set size, a dimension-selection rule that transfers to a second dataset unchanged, and a measured mechanism — Gram-matrix concentration — that predicts where the method stops working. We believe this is a more useful contribution than the one we submitted, and it exists because the reviewers refused the original.

On verification

Every number in the manuscript and in this letter is reproducible from the released repository. Four audit scripts are provided, and we invite the reviewers to run them:

Script	Result	What it checks
<code>audit_c4.py</code>	100/100	Recomputes every published statistic from the raw per-run records using an independent implementation, not the pipeline’s own functions
<code>audit_figures.py</code>	39/39	Every figure is newer than its inputs, and every plotted number is re-derived from the artifacts
<code>audit_prose.py</code>	115/115	Every number written in the manuscript text <i>and in this letter</i> is re-derived from the artifacts
<code>verify_lemma1.py</code>	15/15	The corrected kernel expansion, and the failure of the submitted one

These audits found four genuine errors in our own revision code before publication, including one that produced $n^* = 5$ instead of $n^* = 4$.

1. Associate Editor

AE-1 Advantage claims too generic; evidence limited to few methods and a small numerical improvement

“the reviewers criticize too generic advantage claims versus evidence on a limited set of methods and by a small numerical improvement”

Response. We accept this without reservation. The claim of a general advantage has been withdrawn. The paper is reframed from “where is there an advantage” to “where is a quantum kernel worth its cost”, and the answer is reported as a regime map in which negative and inconclusive verdicts appear on the same footing as positive ones: 21 / 21 / 68 over 110 comparisons.

Changes. New Sec. VI (Regime Map), Fig. 10, and a rewritten abstract and introduction. The introduction states the withdrawn claims on its first page.

AE-2 Insufficient literature coverage, including non-existent references

Response. Both citation defects reported by Reviewer 2 were checked independently and both were confirmed; see R2-5 and R2-6. Four recent studies have been added and the positioning table rewritten.

Changes. New Table I (positioning against four benchmarking studies), Sec. II rewritten, bibliography corrected.

AE-3 Reviewer 1 requests validation on an additional dataset

Response. UNSW-NB15 has been added under the identical protocol. This is what allows us to separate findings that transfer from findings that do not: the selection rule and the entanglement ablation transfer; the sample-complexity reversal does not.

Changes. Sec. V-F, Fig. 11.

AE-4 Comparison with non-SVM methods as a prerequisite for practical advantage claims

Response. A random forest and XGBoost were added to every experiment, tuned by the same procedure as every other model. They change the headline: XGBoost attains the highest mean macro- F_1 at the main operating point.

Changes. All results sections; Figs. 7–12.

AE-5 Reviewer 4 identified issues in the theory

Response. Confirmed and acted on. Theorem 1 was false and has been removed with the objective it maximised; the Pareto stage filtered nothing and has been replaced by a lexicographic rule with no weights. Two further theory defects that no reviewer raised are reported in Sec. 6.

Changes. Sec. III rewritten, including an erratum subsection; new Appendix A.

AE-6 “NISQ-ready” not supported by experiments or noisy simulations

Response. The pipeline is re-run under a noise model derived from the calibration data of a real superconducting backend (**FakeManilaV2**) carrying gate errors, readout errors and thermal relaxation. We are explicit that this is *not* hardware validation: the circuits were not executed on a quantum processor, and the manuscript now says so in those words.

Changes. Sec. V-B and Sec. VII-A, which separates what we measure from what we do not.

2. Reviewer 1

We are grateful for a review that identified the single methodological flaw most damaging to our own conclusions (R1-3), and asked the question whose answer became the paper’s main result (R1-7).

R1-1 Missing QMI 2026 benchmark; Table I needs 2025–2026 literature

“an important recent work is missing: ‘Benchmarking quantum machine learning methods for intrusion detection on noisy quantum computers’ (Quantum Machine Intelligence, 2026). This work should be discussed explicitly”

Response. The work — Cirillo, Esposito and Seo, *Quantum Machine Intelligence*, vol. 8, no. 1, art. 27, 2026 — is now cited, read in full, and discussed in Sec. II-B, with a row for it in Table I. We are glad the reviewer insisted: it is the closest study to ours that we are aware of, and reading it changed what we say.

Two of its findings corroborate ours from a different direction, and we now say so explicitly. It compares against SVM, XGBoost, random forest and a neural network at matched dimensionality and concludes that the classical models retain higher peak accuracy while the quantum models remain competitive in low-dimensional feature spaces — the same ordering we now report, with XGBoost top at our operating point. And it sweeps the qubit count from 2 to 10 and plots F_1 against two-qubit gate count, locating good performance in a 20–35 CNOT band with degradation beyond roughly 60 CNOTs. Our operating point is 24 CNOTs, inside that band, and the collapse we report at 112 CNOTs is consistent with their threshold.

The two studies diverge on the mechanism, and we think that divergence is useful. They attribute the degradation to two-qubit gate error accumulating on noisy backends. We observe the same collapse under exact, noiseless statevector simulation, where no gate error exists, and trace it to concentration of the Gram matrix instead. Hardware noise is therefore not a necessary ingredient; the bound is intrinsic to the kernel.

What that study does not vary — and what, to our knowledge, no study in this area varies — is the training-set size. It is fixed there at a single 15% class-balanced subsample under a random 80/20 split rather than the official partitions. That is the axis on which our contribution lies.

Changes. Sec. II-B; Table I now covers Bowles *et al.* (2024), Schnabel and Roth (2024), the QMI 2026 benchmark, Carducci (2026), and the concurrent study of Gillani *et al.* (2026).

R1-2 Reliance on NSL-KDD; UNSW supplementary was not available

Response. UNSW-NB15 is now a first-class part of the study rather than supplementary material, run under the same protocol with the same audits. Sec. VII-D also quantifies a property of both

corpora that constrains what either can show: NSL-KDD has 610 test rows identical to a training row in both features and label, and UNSW-NB15 is far more heavily duplicated (47.3% of training rows are internal repeats; 25.0% of test rows have an exact duplicate in training).

Changes. Sec. IV-A, Sec. V-F, Sec. VII-D, Fig. 11.

R1-3 Asymmetric hyper-parameter protocol: QSVM fixed at $C = 1.0$, classical SVMs validation-selected

“The hyperparameter selection protocol is asymmetric ... Without such analysis, it is difficult to determine whether the reported performance differences are influenced by the selected regularization parameter.”

Response. The reviewer is right, and the defect ran in our favour. The submitted version described fixing the quantum regulariser as avoiding bias; it does the opposite, granting one family a search the other is denied. Every model, quantum and classical, is now tuned by the same procedure on the same disjoint tuning set of 2,000 records.

We report two arms rather than one: *tuned per N* , in which the search is repeated at each training-set size, and *frozen*, in which hyper-parameters are held at the values selected once. Reporting both is what lets us show that the paper’s main new result is not an artefact of re-tuning (see R1-7).

Changes. Sec. IV-B; Table II reports both arms across all sizes.

R1-4 Conclusions overstated; cannot claim general quantum advantage

Response. Withdrawn. See the summary table at the head of this letter. The abstract now leads with the question rather than with a winning number, and reports 21 / 21 / 68.

Changes. Abstract, Sec. I, Sec. VI, Sec. VIII.

R1-5 Include XGBoost, CatBoost, TabNet, FT-Transformer, or soften practical claims

Response. We added a random forest and XGBoost, and we did *not* add CatBoost, TabNet or FT-Transformer. We prefer to state that plainly rather than imply full coverage.

CatBoost would supply a second gradient-boosting representative without changing the comparison structure. The deep tabular family would require its own tuning-budget protocol to be compared fairly, which would enlarge the study from a controlled quantum-kernel benchmark into a survey of tabular architectures. Accordingly we have softened the practical claims as the reviewer’s alternative allows, and Sec. VII-C states explicitly that we do not claim coverage of all strong tabular methods.

Changes. All experiments; Sec. VII-C.

R1-6 NISQ evidence covers only finite-shot error

Response. Addressed with a backend-derived noise model; see AE-6. Circuits are transpiled onto the backend’s coupling map and basis set, giving depth 59 and 44 two-qubit gates at $n = 4$, $r = 2$. The entanglement conclusion is unchanged under noise.

Changes. Sec. V-B, Sec. VII-A.

R1-7 Is there a crossover point beyond which the classical SVM becomes preferable?

“It would therefore be useful to identify whether there exists a crossover point beyond which the classical SVM becomes preferable, or clarify why such a point is not observed.”

Response. There is a crossover, and it runs in the direction opposite to the one the question anticipates: the classical baselines are preferable at *small* N , and the quantum kernel becomes preferable at large N . Extending the sweep from $N = 1000$ to $N = 10^4$ under the natural class prior:

- At $N = 100$ the ranking is XGBoost 0.7802, random forest 0.7695, QSVM-Z 0.7613, SVM-RBF 0.7296, linear 0.7261, QSVM-ZZ 0.6989.
- At $N = 10^4$ it is QSVM-ZZ 0.7855, QSVM-Z 0.7787, SVM-RBF 0.7740, random forest 0.7728, XGBoost 0.7706.
- The paired difference against XGBoost passes through zero between $N = 2000$ and $N = 5000$ ($-0.0129 \rightarrow +0.0100$) and is Holm-corrected-significant at $N = 5000$ ($p = 0.027$, $d_z = 1.22$) and $N = 10^4$ ($+0.0149$, $p = 0.0078$).

Three qualifications are stated in the manuscript and we repeat them here rather than let the result be read as more than it is. First, the reversal survives freezing the hyper-parameters: all six baseline–arm combinations change sign in the same interval, so it is not a re-tuning artefact. Second, it is not uniform across baselines — against SVM-RBF the difference turns positive but no cell reaches corrected significance, so the quantum kernel never demonstrably beats an RBF kernel at any size we tested. Third, it is a property of the natural prior: repeating the sweep with rare attacks enriched twelvefold to 10% — the sampling our own submitted protocol used — leaves 26 of 30 comparisons inconclusive with no crossover in range.

That third point is the reason the submitted version reported the opposite conclusion, and we regard identifying it as the most useful thing this revision does.

Changes. Sec. V-D, Fig. 9, Table II.

R1-8 Table VI ($N = 1000$) inconsistent with Table IV

Response. Confirmed, and the discrepancy decomposes into two factors that the submitted version conflated. Holding $N = 1000$ fixed and varying only these:

Representation	Test set	QSVM-ZZ	XGBoost
frozen	300-sample subset	0.8469	0.8503
frozen	full KDDTest ⁺	0.7959	0.8043
refit per N	300-sample subset	0.8526	0.8666
refit per N	full KDDTest ⁺	0.8072	0.8242

The test set accounts for about -0.051 macro- F_1 and the representation mode for about $+0.006$. The revision reports both test sets everywhere and states which is which.

Changes. Sec. IV-B; c4_table_iv_vs_vi.csv in the repository.

R1-9 Classical F_1 low relative to values commonly reported on NSL-KDD

Response. The gap is the evaluation protocol, not the models, and it is large enough to be worth stating explicitly. Using the full 122 features and the entire training partition:

Protocol	Random forest	XGBoost
Official split, evaluate on KDDTest ⁺	0.7765	0.8041
Random split <i>within</i> KDDTrain ⁺	0.9978	0.9993

Studies reporting ≈ 0.99 on NSL-KDD are, in the cases we could check, splitting randomly within the training partition rather than using the official train/test partitions, which places near-duplicate records on both sides. We use the official split throughout. This is now stated in the manuscript so that our absolute numbers are not read as underperformance.

Changes. Sec. IV-A; `c4_protocol_vs_literature.csv` in the repository.

R1-10 Supplementary material was not accessible

Response. The supplementary material is replaced by a public repository containing the complete implementation, the frozen protocol files, every intermediate artifact, all figure-generation scripts, and the four audit scripts listed above. The link and the commit hash are given in the Reproducibility paragraph of Sec. IV-D.

Changes. Sec. IV-D.

3. Reviewer 2

R2-1 Baselines limited to SVM variants

Response. Random forest and XGBoost added throughout; see AE-4 and R1-5. The reviewer’s framing — that the real competitor in industry is rarely an SVM — is correct and is now reflected in the result: at the main operating point XGBoost, not the quantum kernel, is the top-ranked model.

Changes. All results sections.

R2-2 Barren plateaus have a kernel analogue: exponential concentration of the kernel matrix

Response. Named, cited, and — we hope usefully — measured. Exponential concentration is now stated as background fact (F3) with the reference the reviewer intends, and it is also the mechanism behind the paper’s sharpest new boundary. Fitting $s(n) \propto n^{-\alpha}$ to the off-diagonal dispersion over $n = 4, \dots, 10$ gives $\alpha_{ZZ} = 0.590$ against $\alpha_Z = 0.292$ at $K = 20$, a ratio of 2.02 — matching the $\binom{n}{2}$ versus n growth in the number of phase terms. Across the same widths the dispersion correlates with macro- F_1 at $r = +0.77$ for the ZZ kernel and $r = +0.32$ for the Z kernel.

Changes. Sec. III (F3), Sec. V-G, Fig. 12, Sec. VII-B.

R2-3 $N_{\text{train}} = 1000$ with five seeds is a thin statistical base

Response. Accepted. All experiments now use ten runs with fixed seeds. Every comparison is paired within run and reported with a t -based 95% interval, a Wilcoxon signed-rank test, the paired effect size d_z , and the fraction of runs with a positive difference; p -values are Holm-corrected within baseline families. Training-set size is now a swept variable rather than a fixed choice, from $N = 100$ to $N = 10^4$.

We also removed a statistical defect the reviewer did not see: the submitted version certified the entanglement effect with a two-sample z -test that treated two arms sharing a training subset as independent, discarding the pairing. The revision compares within run.

Changes. Sec. IV-B, Sec. IV-C, Sec. III-D.

R2-4 Regime map oversells: positive regimes get full statistics, negative regimes only qualitative treatment

“three positive regimes are given full statistical treatment while the two negative regimes get comparatively brief qualitative explanation ... without a matching effect-size/CI treatment.”

Response. Corrected, and the correction changed the balance of the paper. All 110 cells now carry the same five quantities — Δ , 95% CI, Holm-corrected p , d_z , and the positive-run fraction — whether the verdict is favourable to the quantum kernel or against it. The result is 21 favourable, 21 against, 68 inconclusive; the negative regimes, given equal treatment, turn out to be as numerous as the positive ones.

The clearest of them: under feature perturbation, six of six comparisons favour the classical model with effect sizes beyond 2.8. The manuscript presents this as the paper’s clearest negative result rather than in a closing caveat.

Changes. Sec. V-C, Sec. VI, Fig. 10; `regime_map_rows.csv`, all 110 rows independently re-derived by `audit_c4.py`.

R2-5 Reference [15]: article number 116990F should be 116990B

Response. Verified independently before correcting: the SPIE Digital Library URL contains /116990B/, the NASA ADS bibcode is 2021SPIE11699E..0BP, and DOI 10.1117/12.2593297 matches. Corrected.

Changes. Bibliography, entry [15].

R2-6 Reference [26] could not be located; looks fabricated or badly misattributed

Response. We searched twice and independently — by author and title, and by volume and page range — and could not locate the item either. We have removed it, and we record the removal in the released bibliography file together with the original string, so that the change is auditable rather than silent.

Rather than simply delete the sentence it supported, we replaced the citation with a survey that does exist and that we verified before using: A. Kaissar, A. B. Nassif and A. Bouridane, “Enhancing network intrusion detection with quantum machine learning: A comprehensive survey of methods, metrics, and applications,” *Future Internet*, vol. 18, no. 5, art. 234, 2026, doi: 10.3390/fi18050234. It fills the role the withdrawn reference was serving and, being a 2026 survey, also addresses part of AE-2. Two of its findings are relevant enough to state in our Sec. II-B: most published QML–IDS approaches rely on simulated quantum environments and legacy datasets, and evaluation practice is inconsistent across studies.

Changes. Bibliography; Sec. II-B.

4. Reviewer 3

Reviewer 3 recommends rejection on grounds of limited technical novelty. We take the criticism seriously and do not dispute its factual basis: this paper introduces no new quantum kernel, feature map, training method, hardware implementation, or theorem establishing a new regime of advantage. We respond by saying what we think the contribution is instead, and by reporting what changed.

R3-1 The model is a standard ZZFeatureMap configuration; technical novelty is limited

“it does not introduce a new quantum kernel, a new feature map, a new training method, a hardware implementation, or a theoretical result establishing a new regime of advantage.”

Response. Correct as stated, and we have not attempted to manufacture novelty of apparatus. We offer three things instead, and ask that they be weighed on their own terms.

An ordering that reverses with sample size. This is not a re-measurement of a known quantity. To our knowledge no study of quantum kernels on tabular data sweeps the training-set size, and doing so shows that the answer to “does the quantum kernel win” is not a fact about the kernel but a function of N : second-worst of seven models at $N = 100$, best at $N = 10^4$, reversing in a narrow interval, in both tuning arms. A benchmark conducted at a single N — as every study we are aware of is, including our own submitted version — can report either outcome and be internally consistent.

A selection rule that transfers. The three-stage rule returns $n^* = 4$ on NSL-KDD and, with no threshold changed, $n^* = 6$ on UNSW-NB15, recovered on 10/10 independent subsets. The submitted version presented $n = 4$ as a result; it is properly a rule whose output depends on the data.

We would add that the question itself is being treated as a legitimate research question elsewhere. The study Reviewer 4 pointed us to — Carducci, ICAD 2026 — opens by arguing that malware analysts “do not need another general claim that quantum computing is either broadly superior or broadly impractical” but rather “need to know which kinds of malware representations justify quantum effort”, and answers it by locating a bounded intermediate window. We reached the same shape of answer independently, on a different task and with a different primitive. That two studies in adjacent domains converge on “bounded window” rather than on “advantage” is, we think, a result about the technology rather than a shortfall of either paper.

A measured mechanism with a quantitative form. This is the closest we can honestly come to the reviewer’s “theoretical result establishing a regime”, and we present it as an empirical regularity rather than a theorem. The off-diagonal dispersion of the Gram matrix decays as $n^{-\alpha}$ with $\alpha_{ZZ}/\alpha_Z = 2.02$ — the ratio predicted by counting $\binom{n}{2}$ pairwise phase terms against n single-qubit ones — and it predicts end-task macro- F_1 across widths at $r = +0.77$. It tells us where the method stops working, and that boundary is the part of this paper most likely to hold beyond these two datasets.

Changes. Sec. I, Sec. II-D, Sec. V-D, Sec. V-F, Sec. V-G.

R3-2 The advantage rests on a marginal improvement (0.854 vs 0.838)

Response. The claim has been withdrawn, and the specific comparison the reviewer names no longer holds in our own numbers: with a tree ensemble included, XGBoost attains 0.8503 against QSVM-ZZ’s 0.8469, with overlapping intervals. We report this as an ordering of point estimates, not a separation. The reviewer’s underlying point — that a difference of this size does not support an advantage claim — is one the revision now makes itself, quantitatively: with ten runs and Holm correction, the smallest difference reaching any verdict in the map is 0.010 macro- F_1 , and differences as large as 0.031 remain inconclusive.

Changes. Sec. V-B, Sec. VI.

R3-3 The propositions are well-established facts and their framing as proved propositions is inappropriate

Response. Agreed, and corrected. Propositions 1, 3 and 4 are now background facts (F1)–(F3) with citations and no proofs. Proposition 2, which is specific to the comparison this paper makes, is retained as Lemma 1 — and, as Sec. 6 reports, we found in the course of doing so that its stated coefficients were wrong. It is restated correctly with a full proof in Appendix A.

Changes. Sec. III-B, Appendix A, erratum subsection.

R3-4 All experiments use very few qubits and small data, yet are framed as NISQ-aware

Response. Three changes, and one thing we did not do.

We extended the width: the study now includes a full re-run at $K = 80$, where the selection rule requires $n^* = 8$ qubits and 112 two-qubit gates, and the UNSW-NB15 analysis operates at six qubits. The result is negative and we report it as the paper’s sharpest boundary: at eight qubits all 48 paired comparisons favour the classical model, and the entanglement ablation reverses sign. Widening the circuit does not extend the advantage; it removes it, for a reason we can measure in the Gram matrix.

We extended the data: training-set size now runs to $N = 10^4$ with results reported on the complete 22,544-row test partition.

We added a backend-derived noise model with gate, readout and relaxation errors.

What we did not do is execute on a quantum processor. Rather than leave “NISQ-aware” ambiguous, the manuscript now defines it: a statement about circuit budget — qubit count, depth and two-qubit gate count chosen under an explicit cost model — and not a claim of hardware validation. Sec. VII-A states which effects are consequently outside our evidence. The execution path for a real backend is in the released code and has been verified in dry run; we report no hardware numbers and make no claim about them.

Changes. Sec. V-G, Fig. 12, Sec. VII-A, Assumption 1.

R3-5 Results are already established in prior benchmarking studies (arXiv:2403.07059, arXiv:2409.04406)

Response. We have read both and cite both, and our aggregate finding agrees with theirs: general-purpose feature maps rarely beat well-tuned classical baselines. Table I positions this work against them explicitly.

We would note, respectfully, that neither of those studies introduces a new kernel, feature map, training method, hardware implementation, or theoretical result either; both are benchmarks. We raise this not to argue that they should not have been published — we think they should — but because the criterion in R3-1, applied uniformly, would exclude them, which suggests the criterion for a benchmark is not novelty of apparatus but whether it changes what the field believes.

On that criterion the relevant question is what our study does that theirs does not. We read all four in full before answering, and Table I is built from those full texts rather than from abstracts. Bowles *et al.* vary model family across 160 mostly synthetic datasets at up to 18 qubits; Schnabel and Roth vary kernel type across 64 datasets at up to 15 qubits; Cirillo *et al.* vary qubit count from 2 to 10 and plot performance against gate count; Gillani *et al.* run a 4/6/8/12-qubit ablation. Circuit width is therefore *not* an axis we can claim as ours, and we no longer do.

None of the four varies the training-set size. Each fixes it at a single value: fixed per-task subsamples in the first, $M = 240$ in the second, a 15% class-balanced subsample in the third, the full training partition in the fourth. That is the variable that determines the answer here, and it is the one axis on which this work is not duplicated. We do not claim breadth otherwise — Gillani *et al.* evaluate four datasets to our two, at eight qubits to our four and six, under a stricter error-rate criterion.

Changes. Sec. II-C, Sec. II-D, Table I.

5. Reviewer 4

Reviewer 4 found a false theorem and a selection step that filters nothing. Both are confirmed. We are grateful, and we have tried to meet the standard the review sets by auditing the rest of the paper to the same depth and reporting what that turned up, including in the one theoretical statement we retained.

R4-1 Related work: Carducci, ICAD 2026, on the same question for malware detection

Response. Cited and discussed, in a subsection we added for it. We are grateful for the pointer: the parallel is closer than we expected, and it changed how we frame our own boundary.

Carducci reports that the measured advantage is confined to an *intermediate* representational window — classical methods remain stronger both below and above it — and that the pattern survives parameter tuning, tested specifically to rule out a favourable-setting artefact. All three features recur in our results on a different task: our favourable regime is bounded from below by training-set size and from above by circuit width; we report both a frozen and a re-tuned arm for exactly the reason that paper re-tunes; and the representations where that work finds quantum

discovery useful are those dominated by local pairwise structure, which is what the $\binom{n}{2}$ pairwise phase terms of the map encode.

We must be clear about the evidentiary status. We were unable to obtain the full text through our institutional access. The comparison rests on the abstract, the opening of the introduction, and the index terms — from which we can establish that the evaluation is a retrieval setting scored by Recall@ k and that kernel methods are among the techniques involved, but not the qubit count nor whether the training-set size is varied. We say exactly this in Sec. II-D, present the parallel as suggestive rather than as evidence, and do not tabulate the study in Table I, whose cells are otherwise read from full texts. If the reviewer has access and finds that our reading misstates the work, we would be glad to be corrected.

Changes. New Sec. II-D; bibliography; Table I caption states the exclusion and why.

R4-2 Theorem 1 appears misstated; and the Pareto step may not filter anything

“It says $\tilde{F}(4) > \tilde{F}(3)$ but the opposite is shown in Table III ... that may mean Pareto might not be doing any filtering at all, so it might not be a contribution after all.”

Response. Both observations are correct.

On the theorem: the proof asserted $\tilde{F}(4) > \tilde{F}(3)$ while Table III of the same submission reports $\tilde{F}(4) = 0.471 < \tilde{F}(3) = 0.628$. With the correct values the conclusion does not follow — at $\alpha = \beta = \gamma = 1/3$ the objective is maximised at $n = 2$ ($J = 0.551$) and decreases monotonically to 0.059 at $n = 10$, exactly as the reviewer observed from the table.

On Pareto: $V(n)$ increases monotonically while $\tilde{F}(n)$ decreases and $Q(n)$ increases, so no candidate dominates any other and every candidate is Pareto-optimal. The step selected nothing.

We removed Theorem 1 and the objective J rather than repairing them, because the second point shows the construction was not doing the work attributed to it. The replacement is lexicographic and has no weights: retain n with $V(n) \geq 0.85$; among those keep n with $\text{KTA}(n) \geq 0.95 \text{KTA}_{\max}$; among those take the smallest $Q(n)$. On NSL-KDD this returns $n^* = 4$; on UNSW-NB15, unchanged, $n^* = 6$.

Changes. Sec. III-C (new Definition 1 replacing Definition 4, Algorithm 1, Eq. (6) and Theorem 1), erratum subsection, Fig. 5 replacing the Pareto frontier figure.

R4-3 Reproducibility is asserted but no machinery is provided

Response. A public repository is now cited by URL and commit hash, containing the implementation, protocol files, all intermediate artifacts, the figure scripts, and four audit scripts. We regard the audits as the substantive part of this response: `audit_c4.py` recomputes every published statistic from raw per-run records using an independent implementation rather than calling the pipeline’s own statistics functions, since using the code that produced a number to check that number lets shared errors through.

Changes. Sec. IV-D.

R4-4 Proposition 3 is cited to another paper but not derived

Response. Correct. The former Proposition 3 restated a known bound and its “proof” added nothing. It is now background fact (F2), cited to the source and used only diagnostically, with no proof.

Changes. Sec. III-B.

R4-5 No numbers are given for the rare-attack subset, so the +6.7-point claim cannot be verified

“I don’t know if I see any number for the rare-attack subset, so it is not possible to verify the paper’s claim that ‘At $N=500$ QSVM-ZZ still leads by $+6.7$ points over SVM-RBF on the rare-attack subset, with a Cohen’s d of $+0.68$ ’”

Response. The claim cannot be verified because it cannot be reproduced. On inspecting the code that produced the submitted results we found that no rare-class metric was computed at any point. We withdraw the claim.

The revision reports rare-class metrics directly, at every training-set size and for every model. At $N = 10^4$ under the natural prior the rare-class F_1 is:

SVM-RBF	QSVM-ZZ	QSVM-Z	XGBoost	Random forest	Poly-2	Linear
0.585	0.507	0.421	0.334	0.331	0.109	0.089

Two things follow, and we report both. The quantum kernel detects rare attacks far better than either tree ensemble, which the macro average conceals and which matters operationally. But SVM-RBF is better still, by 0.078 — so the honest statement, and the one the manuscript makes, is that an RBF-kernel SVM is the strongest rare-attack detector in this study. Signed per-sample decision margins are also reported, as the submitted version’s Cohen’s d claim implied but did not deliver.

Changes. Sec. V-E; `c4_rare_attack_natural.csv`.

R4-6 The quantum kernel’s hyper-parameters were not tuned

Response. They are now, by the identical procedure applied to every classical model; see R1-3. Because the quantum kernel does not depend on C , its grid costs only repeated fits on an already-computed Gram matrix, so there was never a computational reason for the asymmetry.

Changes. Sec. IV-B.

6. Corrections we identified ourselves

Auditing the manuscript to the standard Reviewer 4’s comments set turned up three further defects that no reviewer raised. We report them because a reviewer comparing the manuscript against the released artifacts would find them, and because two of them are of the same kind as the errors that were raised.

(a) Table III mislabelled its third column. The caption defined $\tilde{F}(n) = 1/\text{DBI}(n)$, but the tabulated values are the ANOVA F statistic; the inverse Davies–Bouldin index takes the values 1.143, 0.982 and 0.922 at $n = 2, 3, 4$. The same column carried a third, also incorrect, label in an earlier draft. No conclusion in the revision depends on the quantity, since it has been removed along with J , but the mislabelling was carried across versions and we record it.

(b) The former Proposition 2 had incorrect coefficients. This is the one theoretical statement the revision retains, and it was wrong. It asserted

$$K^Q = 1 - r^2 \sum_i \Delta_i^2 - r^2 \sum_{i < j} \Delta \varphi_{ij}^2 + O(\|\cdot\|^4) \quad \text{for all } r.$$

Two things are wrong. The coefficients are not equal: the correct weights are 1 and $\frac{1}{4}$, which follows from the gate decomposition, since $P(2x_i) = \exp(-ix_i Z_i)$ up to global phase while the CNOT- $R_Z(\varphi_{ij})$ -CNOT trio realises $\exp(-i\frac{\varphi_{ij}}{2} Z_i Z_j)$. And the r^2 prefactor does not hold at all: for $r \geq 2$ the interleaved Hadamard layers make the relevant operator non-diagonal, and no coefficient pair makes the two-invariant form correct.

Numerically, against an exact statevector kernel verified against Qiskit to 1.3×10^{-15} : at $r = 1$ a least-squares fit recovers exactly 1 and $\frac{1}{4}$ with $R^2 = 1.000000$ at $n = 3, 4, 6$, and the corrected

expansion’s residual decays as $\|\cdot\|^{3.9}$, reaching 5×10^{-11} at $\|\cdot\| = 10^{-3}$; with the submitted coefficients the residual decays only as $\|\cdot\|^{2.0}$, so the claimed second-order accuracy is not attained. At $r = 2$ — the operating point of this study — the two-invariant form attains only $R^2 = 0.60$ to 0.94 with coefficients that change sign between expansion points, while the unrestricted quadratic form still attains $R^2 = 0.999996$.

The lemma is restated for $r = 1$ with the correct coefficients and proved in full in Appendix A, with an explicit remark on why it does not extend. The substantive consequence drawn from it — that SVM-Poly2 is the sharpest classical comparator — is unaffected, since it depends on which monomials appear and not on their coefficients. The verification is released as `verify_lemma1.py`.

(c) The training subsets used in the submitted version were rare-enriched. The per-run training frames contain 10% rare-attack records against a natural rate of 0.83% — an enrichment of roughly twelvefold that was not stated in the submitted manuscript. This is the direct cause of the withdrawn low-data claim: under enrichment there is no crossover, and under the natural prior there is. Both sampling regimes are now reported side by side and the enrichment is stated in the protocol.

7. Note on the marked-up manuscript

Reviewers and the Associate Editor commonly receive a version with changes marked. We would ordinarily supply one, but in this case a marked version would be misleading rather than helpful: essentially every paragraph of the manuscript is new text. The abstract, introduction, related work, framework, experimental setup, results, limitations and conclusion have all been rewritten; the theory section has had one theorem, one definition, one algorithm and one equation removed and a new definition, lemma, remark and appendix added; and all twelve figures are regenerated from re-run experiments. Marking everything would convey no information.

We have therefore prepared this letter to serve that purpose instead: every substantive change is listed above under the comment that motivated it, with the section it lands in. Should the editors prefer a conventionally marked file, we will produce one on request.

Two things did *not* change, and we note them so their absence from the above is not read as an oversight: the title, and the author list and their affiliations.

8. Summary

	AE	R1	R2	R3	R4	Total
Items raised	6	10	6	5	6	33
Addressed with new results	6	9	6	4	5	30
Addressed in part, stated as such	0	1	0	1	1	3

The three items we describe as addressed only in part, and what remains outstanding in each:

- **R4-1** — Carducci (2026) is cited and positioned, but we were unable to obtain the full text through our institutional access, so our discussion of it rests on the abstract and the bibliographic record.
- **R1-5** — we added a random forest and XGBoost, and deliberately did not add CatBoost, TabNet or FT-Transformer. We took the alternative the comment offers and softened the practical claims accordingly.
- **R3-4** — we extended the study to eight qubits, added a backend-derived noise model, and defined what “NISQ-aware” claims. We did not execute on a quantum processor, and we make no hardware claim.

We thank the reviewers again. The paper reports a narrower and partly negative result than the one we submitted, and it is a better paper for it.

Sincerely,

The authors

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